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HOUSE OF ASPIRANTS

COMPUTER

CHAPTER 13

Modulation and Demodulation

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WEBSITE
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EDITION
October 2026

LANGUAGE
Punjabi & English

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CHAPTER 13

Modulation and Demodulation

Computer Awareness

IN THIS CHAPTER

- Modulation and demodulation
- Signal characteristics
- AM, FM and PM
- Frequency bands
- Bandwidth
- Attenuation, distortion and noise

- Infrared = TV Remote.
- Microwaves = Line of Sight.
- Satellite = Global Communication.

MOST IMPORTANT FACTS FOR EXAMS

1.Client-Server Architecture is centralized.

2.Peer-to-Peer Architecture is decentralized.

3.Hybrid combines both architectures.

4.Twisted Pair Cable is the most commonly used networking cable.

5.UTP has no shielding.

6.STP has metallic shielding.

7.Coaxial Cable is used in Cable TV.

8.Fiber Optic uses light signals.

9.Throughput means actual transmission rate.

10.Parity Bit detects transmission errors.

11.Radio Waves are omnidirectional.

12.Infrared is used in TV remotes.

13.Microwaves require line-of-sight communication.

14.Satellite communication provides worldwide coverage.

MODULATION AND DEMODULATION

Communication systems transmit information over a communication medium using signals. Before transmission, data must be converted into a suitable signal format.

MODULATION

Modulation changes a carrier signal according to an information signal so that the information can be transmitted.

Modulation means adding information to a carrier signal so that it can travel over long distances.

Example

When you speak on a mobile phone, your voice is converted into electrical signals and modulated before transmission.

DEMODULATION

Demodulation is the process of extracting the original information from the received modulated signal.

Demodulation means recovering the original data from the received signal.

Example

When the receiver hears your voice during a phone call, the received signal is demodulated back into sound.

MODULATION VS DEMODULATION

Feature	Modulation	Demodulation
Meaning	Adding information to carrier signal	Extracting information from signal
Location	Sender Side	Receiver Side
Purpose	Transmission	Recovery of Data

MODEM (MOdulator DEModulator)

A Modem is a communication device that performs both:

- Modulation
- Demodulation

CARRIER SIGNAL

A Carrier Signal is a signal that contains no information but has characteristics that can carry information.

Characteristics of Carrier Signal

1.Amplitude

2.Frequency

3.Phase

MODULATED SIGNAL

A Modulated Signal is formed by combining:

- Carrier Signal
- Information (Message) Signal

Formula

Modulated signal = Carrier with a parameter changed by the information signal; this is not generally simple signal addition.

Exam Point

Modulated Signal carries actual information.

CHARACTERISTICS OF A SIGNAL

A signal is mainly characterized by:

Characteristic	Meaning	Unit
Amplitude	Strength of Signal	Volt (V)
Frequency	Number of Cycles per Second	Hertz (Hz)
Phase	Position of Wave	Degrees or radians

1. Amplitude

Amplitude is the maximum height or strength of a signal.

Unit :Volt (V)

Example :Higher amplitude means stronger signal strength.

2. Frequency

Frequency is the number of cycles completed by a signal in one second.

Unit :Hertz (Hz)

Example :1 Hz = 1 cycle per second

3. Phase

Phase represents the position of a wave relative to a reference point.

Phase is an angle, commonly expressed in degrees or radians.

Exam Point

Amplitude → Volt

Frequency → Hertz

Phase → Degrees or radians

TYPES OF MODULATION

Information can be added to a carrier signal by changing:

- Amplitude

- Frequency
- Phase

1. AMPLITUDE MODULATION (AM) In Amplitude Modulation, the amplitude of the carrier signal changes while frequency and phase remain constant.

Example :AM Radio Broadcasting

2. FREQUENCY MODULATION (FM)

In Frequency Modulation, the frequency of the carrier signal changes while amplitude remains constant.

Example :FM Radio Broadcasting

3. PHASE MODULATION (PM)

In Phase Modulation, the phase of the carrier signal changes according to the information signal.

Characteristics

- Better signal quality
- Better noise resistance
- Higher efficiency

The efficiency of PM, FM and AM depends on bandwidth, noise, power and the design of the communication system; PM is not universally more efficient.

FREQUENCY BANDS

Frequency bands are ranges of frequencies used for communication.

1. RADIO FREQUENCY (RF) BAND

Radio Frequency Band is used for wireless transmission of signals and data.

Uses

- Mobile Communication
- Wi-Fi
- Radio Broadcasting
- Television Broadcasting

2. VOICE BAND

Voice Band is the frequency range used for telephone voice communication.

Uses

- Telephone Calls
- Voice Communication Systems

3. NARROWBAND

A Narrowband communication channel uses a small bandwidth.

Example

Traditional Telephone Networks

Exam Point

Narrowband = Low Bandwidth

4. BROADBAND

Broadband communication uses a wide frequency range and supports high-speed data transmission.

Examples

- Fiber Internet
- Broadband Internet Connections
- Cable Internet

Exam Point Broadband = High Bandwidth + High Speed

FACTORS AFFECTING TRANSMISSION MEDIA

The efficiency of a communication channel depends on several factors.

Transmission Media Factors

- Bandwidth
- Interference
- Transmission Impairment

BANDWIDTH

In networking, bandwidth commonly means the maximum data rate in bits per second. Frequency bandwidth is the width of a frequency band and is measured in hertz.

Units of Bandwidth

Unit	Full Form
bps	Bits per Second
Kbps	Kilobits per Second
Mbps	Megabits per Second
Gbps	Gigabits per Second
Tbps	Terabits per Second

Example

A 100 Mbps Internet connection can transfer more data than a 10 Mbps connection.

Memory Trick

Bandwidth = Road Width

Wider Road → More Vehicles

Higher Bandwidth → More Data

INTERFERENCE

Interference is the disturbance of a signal caused by unwanted external signals.

Effects

- Data Errors
- Signal Quality Reduction
- Communication Failure

Example

Noise from electrical equipment affecting radio communication.

TRANSMISSION IMPAIRMENT

Transmission Impairment occurs when the received signal differs from the transmitted signal.

Result

- Poor signal quality
- Data corruption
- Communication errors

CAUSES OF TRANSMISSION IMPAIRMENT

Transmission Impairment

- Attenuation
- Distortion
- Noise 1. ATTENUATION

Attenuation means loss of signal strength as the signal travels through a transmission medium.

Cause

Signal loses energy while travelling long distances.

Solution

Amplifiers are used to restore signal strength.

Example

Mobile signals become weaker at longer distances.

2. DISTORTION

Distortion is a change in the shape or form of a signal during transmission.

Cause

Different frequency components travel at different speeds and reach the destination at different times.

Example

Poor-quality audio transmission.

3. NOISE

Noise is an unwanted signal that mixes with the original signal.

Types of Noise

1. Thermal Noise

Produced due to heat in electronic devices.

2. Crosstalk Noise

Occurs when signals from one channel interfere with another channel. 3. Induced Noise

Caused by external electromagnetic sources.

4. Impulse Noise

Sudden spikes in voltage caused by lightning or power disturbances.

MOST IMPORTANT FACTS FOR EXAMS

1.Modem stands for Modulator-Demodulator.

2.Carrier signal contains no information.

3.Amplitude is measured in Volts.

4.Frequency is measured in Hertz.

5.Phase is an angle, commonly expressed in degrees or radians.

6.AM changes amplitude.

7.FM changes frequency.

8.PM changes phase.

9.Modulation efficiency depends on the system; PM is not always more efficient than AM or FM.

10.RF Band is used for wireless communication.

11.Broadband provides high-speed data transfer.

12.Bandwidth is measured in bps.

13.Attenuation means signal loss.

14.Distortion means signal shape change.

15.Noise means unwanted signal.

16.Amplifiers compensate for attenuation